Plan and prepare to develop AI solutions on Azure

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1. **What is AI?**
   a. AI solutions are built on machine learning models that encapsulate semantic relationships found in huge quantities of data, en applications to appear to interpret input in various formats, reason over the input data, and generate appropriate responses a predictions.

   b. Common AI Capability:
      i. **Generative AI**: The ability to generate original responses to natural language prompts.
      ii.
      iii. Agents: Generative AI applications that can respond to user input or assess situations autonomously, and take appropriate actions.

      iv. Computer vision: The ability to accept, interpret, and process visual input from images, videos, and live camera streams.

      v. Speech: The ability to recognize and synthesize speech.

      vi. Natural language processing: The ability to process natural language in written or spoken form, analyze it, identify key poi and generate summaries or categorizations.

      vii. Information extraction: The ability to use computer vision, speech, and natural language processing to extract key informa from documents, forms, images, recordings, and other kinds of content.

      viii. Decision support: The ability to use historic data and learned correlations to make predictions that support business decis making.

   c. Generative AI uses language models to respond to natural language prompts, enabling you to build conversational apps and ag that support research, content creation, and task automation in ways that were previously unimaginable.

   d. The language models used in generative AI solutions can be large language models (LLMs) that have been trained on huge volu of data and include many millions of parameters; or they can be small language models (SLMs) that are optimized for specific scenarios with lower overhead.

2. **Foundry Tools:**
   a. Microsoft Azure provides a wide range of cloud services that you can use to develop, deploy, and manage an AI solution.

   b. AI development on Azure is Foundry Tools; a set of out-of-the-box prebuilt APIs and models that you can integrate into your applications.

   c. Commonly used Foundry Tools (Full List: https://learn.microsoft.com/en-us/azure/ai-services/what-are-ai-services#available-a ai-services?azure-portal=true)

      i. **Azure OpenAI**: Azure OpenAI in Foundry Models provides access to OpenAI generative AI models including the GPT family large and small language models and DALL-E image-generation models within a scalable and securable cloud service on Az

      ii. **Azure Vision**: The Azure Vision service provides a set of models and APIs that you can use to implement common comput vision functionality in an application. With the AI Vision service, you can detect common objects in images, generate capti descriptions, and tags based on image contents, and read text in images.

      iii. **Azure Speech**: The Azure Speech service provides APIs that you can use to implement text to speech and speech to text transformation, as well as specialized speech-based capabilities like speaker recognition and translation.

      iv. **Azure Language**: The Azure Language service provides models and APIs that you can use to analyze natural language text perform tasks such as entity extraction, sentiment analysis, and summarization. The AI Language service also provides functionality to help you build conversational language models and question answering solutions.

      v. **Microsoft Foundry Content Safety**: Microsoft Foundry Content Safety provides developers with access to advanced algori for processing images and text and flagging content that is potentially offensive, risky, or otherwise undesirable.

      vi. **Azure Translator**: The Azure Translator service uses state-of-the-art language models to translate text between a large nu of languages.

      vii. **Azure AI Face**: The Azure AI Face service is a specialist computer vision implementation that can detect, analyze, and reco human faces. Because of the potential risks associated with personal identification and misuse of this capability, access to

features of the AI Face service are restricted to approved customers.

viii. **Azure AI Custom Vision**: The Azure AI Custom Vision service enables you to train and use custom computer vision models image classification and object detection.

ix. **Azure Document Intelligence**: With Azure Document Intelligence, you can use pre-built or custom models to extract fields complex documents such as invoices, receipts, and forms.

x. **Azure Content Understanding**: The Azure Content Understanding service provides multi-modal content analysis capabiliti that enable you to build models to extract data from forms and documents, images, videos, and audio streams.

xi. **Azure AI Search**: The Azure AI Search service uses a pipeline of AI skills based on other Foundry Tools and custom code to extract information from content and create a searchable index. AI Search is commonly used to create vector indexes for that can then be used to ground prompts submitted to generative AI language models, such as those provided in Azure OpenAI.

d. To use Foundry Tools, you create one or more Azure AI resources in an Azure subscription and implement code in client applic to consume them. In some cases, AI services include web-based visual interfaces that you can use to configure and test your resources - for example to train a custom image classification model using the Custom Vision service you can use the visual int to upload training images, manage training jobs, and deploy the resulting model.

e. Most Foundry Tools, such as Azure Vision, Azure Language, and so on, can be provisioned as standalone resources, enabling yo create only the Azure resources you specifically need.

f. Each standalone Azure AI resource provides an endpoint and authorization keys that you can use to access it securely from a cl application.

g. Alternatively, you can provision a Foundry Tools resource that encapsulates multiple AI services in a single Azure resource. Usi Foundry Tools resource can make it easier to manage applications that use multiple AI capabilities. There are two Foundry reso types you can use:

i. **Foundry Tools**:
The Foundry Tools resource type includes the following services, making them available from a single endpoint:
Azure Speech
Azure Language
Azure Translator
Azure Vision
Azure AI Face
Azure AI Custom Vision
Azure Document Intelligence

ii. **Microsoft Foundry**:
The Microsoft Foundry resource type includes the following services, and supports working with them through a Mic Foundry project*:
Azure OpenAI
Azure Speech
Azure Language
Microsoft Foundry Content Safety
Azure Translator
Azure Vision
Azure AI Face
Azure Document Intelligence
Azure Content Understanding

h. **Regional availability**:
i. Some services and models are available in only a subset of Azure regions. Consider service availability and any regional qu restrictions for your subscription when provisioning Foundry Tools.
ii. Use the https://azure.microsoft.com/explore/global-infrastructure/products-by-region/table to check regional availabilit Azure services.
iii. Use the https://learn.microsoft.com/en-us/azure/foundry/foundry-models/concepts/models-sold-directly-by-azure?tabs=global-standard-aoai%2Cglobal-standard#model-summary-table-and-region-availability?azure-portal=true in the Azu OpenAI documentation to determine regional availability for Azure OpenAI models.

i. **Cost**:
i. Foundry Tools are charged based on Usage
ii. **Foundry Tool Pricing**: https://azure.microsoft.com/en-us/pricing/details/foundry-tools/ to understand pricing for the AI services you intend to incorporate into your application
iii. **Azure pricing calculator**: https://azure.microsoft.com/en-us/pricing/calculator/ to estimate the costs your expected usage incur.

3. Microsoft Foundry:

a. Microsoft Foundry is a platform for AI development on Microsoft Azure. While you can provision individual Foundry Tools reso and build applications that consume them without it, the project organization, resource management, and AI development capabilities of Microsoft Foundry makes it the recommended way to build all but the most simple solutions.

b. Microsoft Foundry provides the Microsoft Foundry portal, a web-based visual interface for working with AI projects.

c. It also provides the Microsoft Foundry SDK, which you can use to build AI solutions programmatically.

d. Microsoft Foundry projects: In Microsoft Foundry, you manage the resource connections, data, code, and other elements of th solution in projects. There are two kinds of project:

   i. Foundry projects:
      1. Foundry projects are associated with a Microsoft Foundry resource in an Azure subscription. Foundry projects provid support for Microsoft Foundry Models (including OpenAI models), Microsoft Foundry Agent Service, Foundry Tools, a tools for evaluation and responsible AI development.

      2. You can use Microsoft Foundry portal to work in projects that are based in Microsoft Foundry resources, making it ea add connected resources and manage model and agent deployments.

      3. A Microsoft Foundry resource supports the most common AI development tasks to develop generative AI chat apps a agents.

   ii. Hub-based projects (deprecated):
      1. Hub-based projects are associated with an Azure AI hub resource in an Azure subscription. Hub-based projects includ Microsoft Foundry resource, as well as managed compute, support for Prompt Flow development, and connected Az storage and Azure key vault resources for secure data storage. These are now deprecated, but still in use in several projects today.

      2. Azure AI hub resources support advanced AI development scenarios, like developing Prompt Flow based applications fine-tuning models. You can also use Azure AI hub resources in both Microsoft Foundry portal and Azure Machine Lea portal, making it easier to work on collaborative projects that involve data scientists and machine learning specialists well as developers and AI software engineers

e. What is Microsoft Foundry? : https://learn.microsoft.com/en-us/azure/foundry/what-is-foundry

4. Developer tools and SDKs:

a. While you can perform many of the tasks needed to develop an AI solution directly in the Microsoft Foundry portal, developer need to write, test, and deploy code.

b. There are many development tools and environments available, and developers should choose one that supports the language SDKs, and APIs they need to work with and with which they're most comfortable. (E.g. VS Code)

c. When developing Microsoft Foundry based generative AI applications in Visual Studio Code, you can use the Microsoft Foundr Visual Studio Code extension to simplify key tasks in the workflow, including:

   i. Browsing and managing project resources, including deployed models, agents, connections, and vector stores.

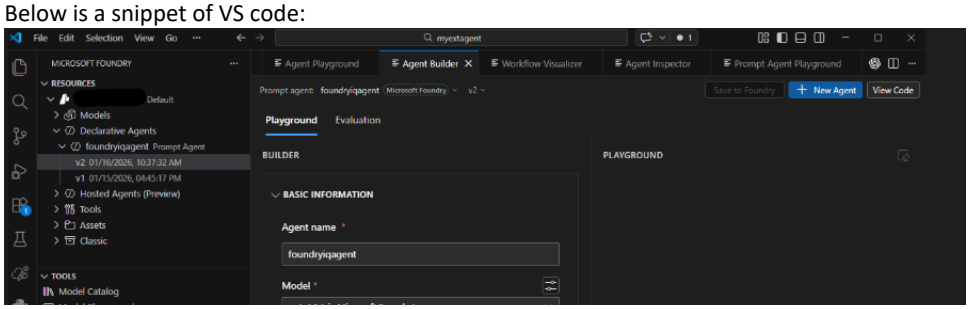
   ii. Deploying models from the model catalog.

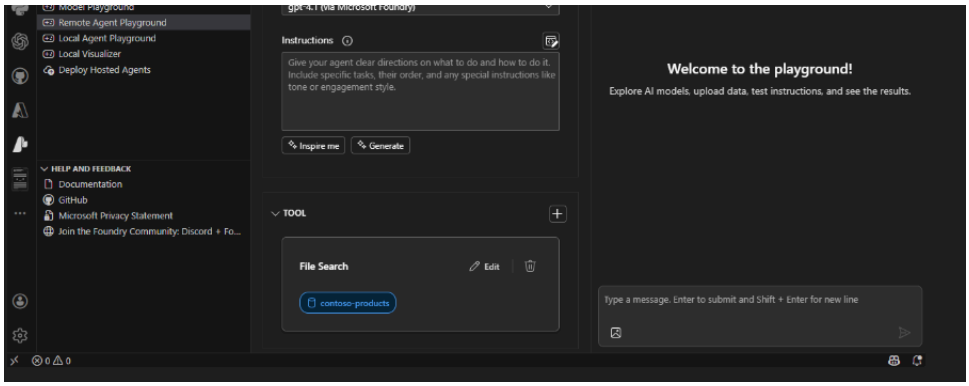
   iii. Testing models and agents in integrated playgrounds.

   iv. Configuring declarative and hosted agents using a visual designer and YAML files.

   v. Generating integration code to connect agents with your applications.

   vi. Below is a snippet of VS code:



vii. 

d. GitHub and GitHub Copilot:
   i. GitHub is the world's most popular platform for source control and DevOps management, and can be a critical element of team development effort. Visual Studio and VS Code both provide native integration with GitHub, and access to GitHub Co an AI assistant that can significantly improve developer productivity and effectiveness.

e. Programming languages, APIs, and SDKs:
   i. When building AI solutions on Azure, some common SDKs you should plan to install and use include:
      1. The Microsoft Foundry SDK, which enables you to write code to connect to Microsoft Foundry projects and access res connections, which you can then work with using service-specific SDKs.

      2. The Microsoft Foundry Models API, which provides an interface for working with generative AI model endpoints host Microsoft Foundry.

      3. The Azure OpenAI in Microsoft Foundry Models API, which enables you to build chat applications based on OpenAI m hosted in Microsoft Foundry.

      4. Foundry Tools SDKs - AI service-specific libraries for multiple programming languages and frameworks that enable yo consume Foundry Tools resources in your subscription. You can also use Foundry Tools through their REST APIs.

      5. The Microsoft Foundry Agent Service, which is accessed through the Microsoft Foundry SDK and can be integrated wi frameworks like Semantic Kernel to build comprehensive AI agent solutions.

   ii. Links:
      1. Microsoft Foundry SDKs and Endpoints: https://learn.microsoft.com/en-us/azure/foundry/how-to/develop/sdk-over pivots=programming-language-python
      2. Azure AI Model Inference REST API reference: https://learn.microsoft.com/en-us/rest/api/aifoundry/modelinference
      3. Azure OpenAI in Microsoft Foundry Models REST API reference: https://learn.microsoft.com/en-us/azure/foundry/openai/reference
      4. Foundry Tools SDK reference: https://learn.microsoft.com/en-us/azure/ai-services/reference/sdk-package-resources pivots=programming-language-csharp
      5. Foundry Tools REST API reference: https://learn.microsoft.com/en-us/azure/ai-services/reference/rest-api-resources
      6. Foundry Agent Service: https://learn.microsoft.com/en-us/azure/foundry/agents/overview
      7. Introduction to Semantic Kernel: https://learn.microsoft.com/en-us/semantic-kernel/overview/

5. **Responsible AI:**
   a. It's important for software engineers to consider the impact of their software on users, and society in general; including considerations for its responsible use.

   b. The potential for harm to individuals or groups through incorrect predictions or misuse of AI capabilities is a major concern, an software engineers building AI-enabled solutions should apply due consideration to mitigate risks and ensure fairness, reliabili and adequate protection from harm or discrimination.

   c. These considerations are particularly important due to the nature of how AI systems work and inform decisions; often based o probabilistic models, which are in turn dependent on the data with which they were trained.

   d. Core Principles for Responsible AI in Microsoft:
      1. Fairness:
         a. AI systems should treat all people fairly. For example, suppose you create a machine learning model to support a approval application for a bank. The model should make predictions of whether or not the loan should be appro without incorporating any bias based on gender, ethnicity, or other factors that might result in an unfair advanta disadvantage to specific groups of applicants.

      2. Reliability and Safety:

a. AI systems should perform reliably and safely. For example, consider an AI-based software system for an autono vehicle; or a machine learning model that diagnoses patient symptoms and recommends prescriptions. Unreliab these kinds of system can result in substantial risk to human life.

3. Privacy and security:
   a. AI systems should be secure and respect privacy. The machine learning models on which AI systems are based re large volumes of data, which may contain personal details that must be kept private

4. Inclusiveness:
   a. AI systems should empower everyone and engage people. AI should bring benefits to all parts of society, regardl physical ability, gender, sexual orientation, ethnicity, or other factors.

5. Transparency:
   a. AI systems should be understandable. Users should be made fully aware of the purpose of the system, how it wo and what limitations may be expected.

6. Accountability:
   a. People should be accountable for AI systems. Although many AI systems seem to operate autonomously, ultimat it's the responsibility of the developers who trained and validated the models they use, and defined the logic tha bases decisions on model predictions to ensure that the overall system meets responsibility requirements.

6. **Exercise**: https://microsoftlearning.github.io/mslearn-ai-studio/Instructions/Exercises/01-Explore-ai-studio.html

7. **Module Assessment:**
   a. Which Azure resource provides language and vision services from a single endpoint?
      i. Foundry Tools
   b. You plan to create a simple chat app that uses a generative AI model. What kind of project should you create?
      i. Microsoft Foundry Project
   c. Which SDK enables you to connect to resources in a project?
      i. Microsoft Foundry SDK